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# Protein Circuit Topology Plugin

## User Manual & Documentation

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### Abstract

This is the official documentation file for the PyMOL Protein Circuit Topology plugin. The plugin wraps most of the functionality of the ProteinCT framework [1] and exposes it through a PyMOL GUI. For questions, suggestions, or bug reports, please contact us via the email address above. The latest version is available at the [GitHub repository](#); the release described in the accompanying SoftwareX article is tag `v0.0.3`.

**Circuit Topology in one paragraph.** Circuit Topology (CT) [2] describes how pairs of residue–residue contacts — within a chain and, for multi-chain objects, between chains — are arranged along the sequence. A *contact pair* is a pair of residues in spatial proximity; a *relation* is the arrangement of two contact pairs. The plugin exposes the four relation types also returned by ProteinCT’s `local_ct`: **Series (S)** — sequentially arranged, non-overlapping contact pairs; **Parallel (P)** — one contact pair nested inside the other; **Inverse parallel (IP)** — inverse nesting; and **Cross (X)** — interleaved contact pairs. A relation is a statement about the order of the four endpoints along the sequence; in particular a Cross relation does not by itself imply a physical crossing, threading or knot. A *contact map* is the set of residue pairs within a distance cutoff; the plugin renders it as a residue  $\times$  residue matrix or as an arc (“circuit”) diagram above the sequence. In a *relations matrix*, every contact-pair cell is coloured by its relation type.

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# 1 Installation Guide

## 1.1 Downloading and Installing PyMOL

Download the PyMOL 3.1.6.1 installer for your platform (Windows / Linux / macOS) from <https://www.pymol.org/>. Previous versions are available at:

<https://storage.googleapis.com/pymol-storage/installers/index.html>

During setup, install PyMOL for the current user only by selecting **Just Me (recommended)**. This ensures the plugin’s automatic dependency installer has the necessary write permissions. If PyMOL is installed for all users, the installer needs administrator or root privileges; the plugin detects this, explains it, and prints the command to run yourself.

When PyMOL prompts you to register file extensions, select **Register Recommended**. The plugin works with `.pdb` and `.cif` structures and with `.xtc` trajectories.

## 1.2 Mandatory Dependencies

Table 1: Required Python and PyMOL versions. These are the versions the released outputs were produced with.

| Dependency | Version | Dependency   | Version |
|------------|---------|--------------|---------|
| Python     | 3.10.18 | NumPy        | 1.26.4  |
| PyQt       | 5.15.11 | Pandas       | 2.3.3   |
| Qt-Main    | 5.15.15 | Matplotlib   | 3.9.1   |
| PyQt5-sip  | 12.17.0 | PyMOL        | 3.1.6.1 |
| Biopython  | 1.85    | PyMOL-Bundle | 3.1.6.1 |

Only four of these are the plugin’s own requirements — NumPy, Pandas, Matplotlib and Biopython — and they are declared in `requirements.yml`. Everything else ships with the PyMOL bundle. On a stock PyMOL 3.1.6.1 installation only `pandas` and `matplotlib` are actually missing, so those are the only two packages the automatic installer normally has to fetch.

## 1.3 Plugin Activation Steps

1. Download the latest plugin release `.zip` directly from GitHub:

[protein-circuit-topology-pymol-plugin.zip](#)

For a specific tagged version, browse the [Releases page](#); the version described in the SoftwareX article is [v0.0.3](#).

2. Open PyMOL and navigate to **Plugin** → **Plugin Manager**.

3. Switch to the **Install New Plugin** tab and click **Choose file...**, then select the .zip you just downloaded.

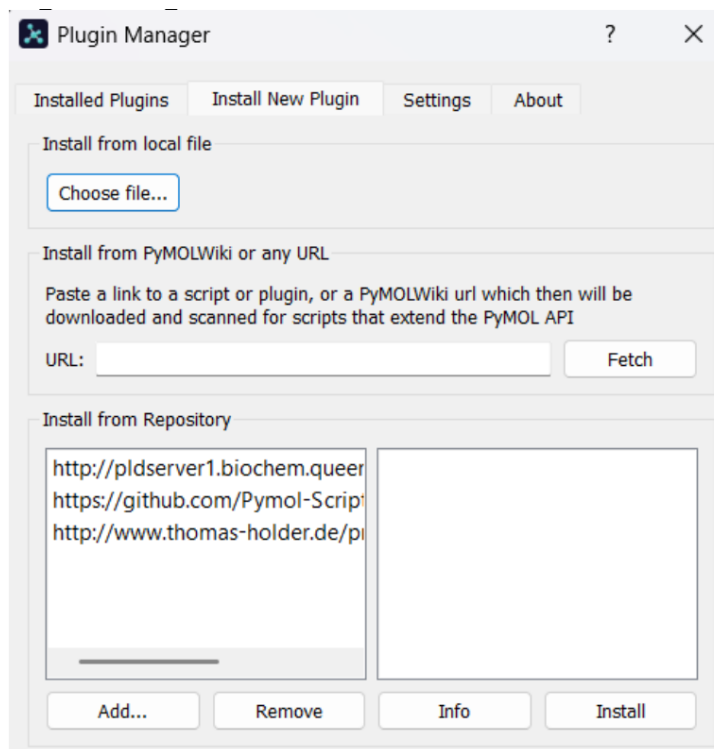


Figure 1: PyMOL's Plugin Manager with the **Install New Plugin** tab open. Use the **Choose file...** button under *Install from local file* to select the downloaded .zip.

4. Wait for the plugin to install and configure its dependencies. After installation, launch the plugin from **Plugin** → **Protein Circuit Topology Plugin**.

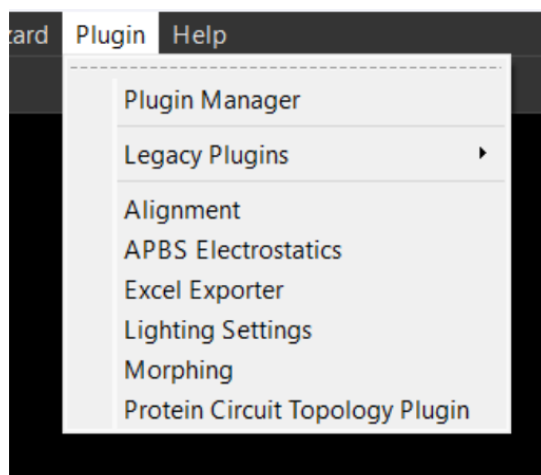


Figure 2: The plugin appears under the PyMOL **Plugin** menu after installation.

### Important Installation Notes

- **Initial setup:** the first time the plugin is loaded, PyMOL may need extra time to download and configure the missing dependencies. Progress is printed to the PyMOL command line.
- **Restart prompt:** if the packages install correctly but Python cannot import them in the already-running session, the plugin says so and asks you to restart PyMOL. Nothing is lost — reopen PyMOL and the plugin is ready.
- **Nothing is replaced:** the installer only *adds* packages. PyMOL and Python are pinned to the versions you already have, so the solver cannot swap them out while resolving the new dependencies.

## 1.4 What the Automatic Installer Does

The installer works entirely through the conda environment that PyMOL itself runs in. It never creates a new environment and never touches any other environment on the system. In outline it

1. locates the conda belonging to the running PyMOL (CONDA\_EXE first, then the usual paths under `sys.prefix`, then `conda` on `PATH`);
2. works out which of the four requirements are missing; and
3. installs only those, with the interpreter and PyMOL pinned:

```
conda install --yes --freeze-installed --prefix <PyMOL's sys.prefix>
python==<installed> pymol==<installed>
conda-forge::<each missing package>
```

The pins matter because they stop conda from satisfying the new dependencies by replacing PyMOL itself.

### Manual Installation

If you would rather install the dependencies yourself, run the equivalent command before loading the plugin. Find the prefix and the versions to pin with

```
python -c "import sys, pymol; print(sys.prefix,
sys.version.split()[0], pymol.cmd.get_version()[0])"
```

and then

```
conda install --yes --freeze-installed --prefix <prefix>
python==<version> pymol==<version>
"conda-forge::numpy">=1.23,<2" "conda-forge::pandas">=2.0,<3"
"conda-forge::matplotlib">=3.7,<4" "conda-forge::biopython">=1.80,<2"
```

## 2 User Guide

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The GUI opens as a single window titled *Circuit Topology Tool* with three tabs: **Local Circuit Topology**, **Single-File Analysis** and **Multi-File Analysis**. Small grey *i* buttons throughout the interface explain individual controls on hover or click.

Non-polymer atoms are handled for you

Circuit Topology analyses polymer atoms only. Waters, ions, ligands and other non-polymer atoms are excluded automatically at analysis time, and nucleic-acid and UNK residues are dropped when the chain is parsed. Nothing is removed from your loaded object, and no cleanup step is required before running an analysis. When you pick an object, the plugin prints to the PyMOL command line exactly what it will ignore, for example:

```
Excluded 118 non-polymer atoms (58 HOH, 2 ZN) from analysis; 1bcs is unchanged.
```

### 2.1 Local Circuit Topology (LCT) Protocol

The Local Circuit Topology tab analyses the topological relationships of one targeted residue against the rest of the structure.

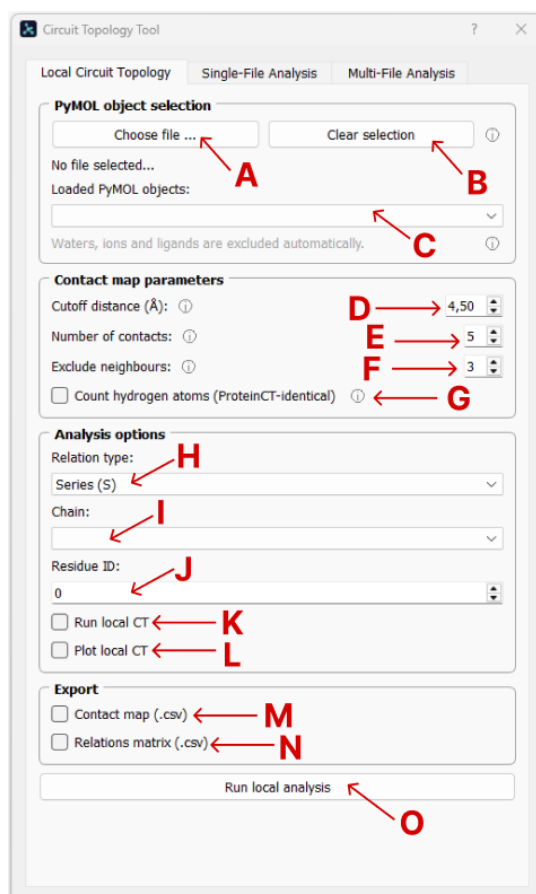


Figure 3: The Local Circuit Topology tab, with the controls used in this protocol labelled A–N: object selection (A–C), contact-map parameters (D) cutoff distance, (E) number of contacts, (F) exclude neighbours (the *Count hydrogen atoms* checkbox sits below (F)); analysis options (G) relation type, (H) chain, (I) residue ID, (J) run local CT, (K) plot local CT; exports (L) contact map, (M) relations matrix; and the run button (N).

To narrow the analysis to a specific region, select a chain (H) and a residue ID (I), then choose the relation type (G) of interest. With **Run local CT** (J) ticked the plugin computes, for every contact pair that begins or ends at the selected residue, how it relates topologically to every other contact pair in the chain. Results can be exported as .csv files via (L) and (M).

## LCT Step-by-Step Guide

1. **Load a structure:** click **Choose file ...** (A) to load a .pdb or .cif file, or pick an already-loaded PyMOL object from the **Loaded PyMOL objects** dropdown (C). **Clear selection** (B) unloads the current file or object from PyMOL again.
2. **Define contact parameters (D–F):**
  - **Cutoff distance (Å) (D)** — maximum atom–atom distance counted as a contact. Range 0.1–20.0, default 4.5.
  - **Number of contacts (E)** — minimum number of atom–atom contacts required before a residue–residue pair counts as a contact. Range 5–45, default 5.

- **Exclude neighbours (F)** — discards trivial contacts between residues close in sequence. Range 1–10, default 3.
  - **Count hydrogen atoms (ProteinCT-identical)** — off by default, so only heavy atoms enter the atom–atom criterion and hydrogen-bearing files (NMR entries, MD frames) give the same contact pairs as their heavy-atom counterparts. Tick it to count every atom present, exactly as the reference ProteinCT implementation does.
3. **Target a region:** pick the **Chain (H)** and the **Residue ID (I)**. The residue spinner is bounded by the residues actually present in the chosen chain, and it is repopulated whenever you change object or chain.
  4. **Choose Relation Type (G):** **Series (S)**, **Parallel (P)**, **Inverse parallel (IP)** or **Cross (X)**.
  5. **Output options:**
    - Tick **Run local CT (J)** to compute the per-residue relation-type counts, which are printed to the PyMOL command line.
    - Tick **Plot local CT (K)** to draw the plots shown below.
    - Tick **Contact map (.csv) (L)** or **Relations matrix (.csv) (M)**. Ticking either one reveals a **Choose output directory ...** button below the checkboxes; pick a destination before running.
  6. **Run:** click **Run local analysis (N)**.

#### When nothing is highlighted

Not every combination of chain, residue and relation type has anything to show. If the selected residue takes part in contact pairs but none of them stand in the requested relation, the plugin says so on the command line, and suggests trying another relation type or residue. If the residue takes part in no contact pairs at all, that is reported too — a common outcome when the cutoff distance is small or **Exclude neighbours** is large.

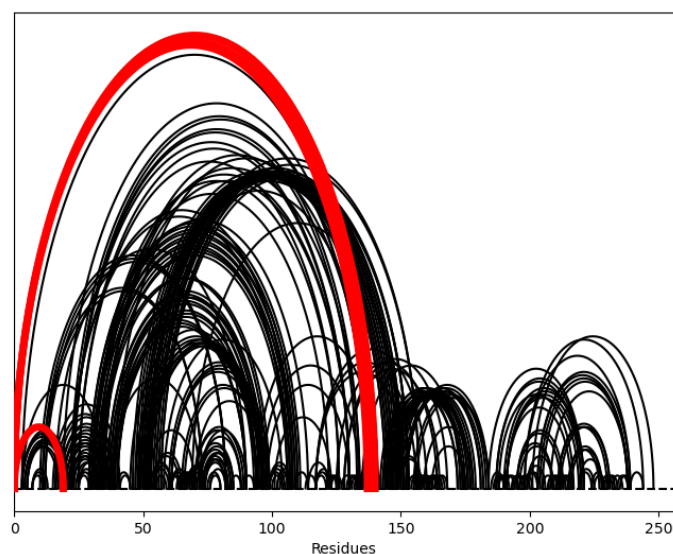


Figure 4: Local CT plot, step 1: contact pairs that begin or end at the selected residue are highlighted in red over the background contact map.

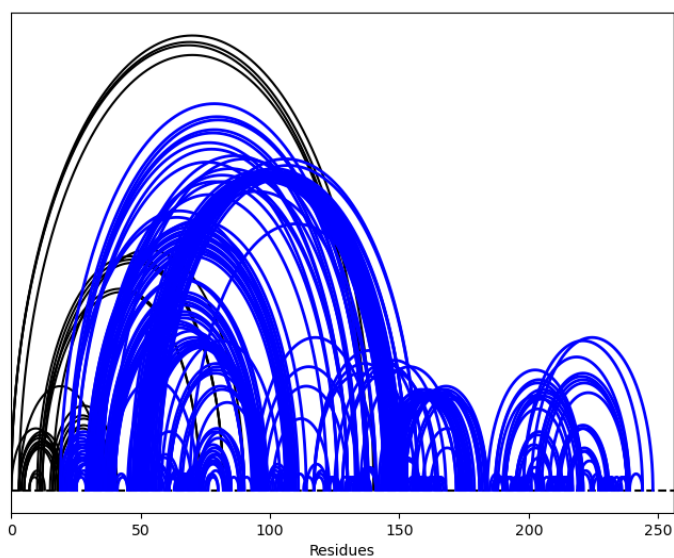


Figure 5: Local CT plot, step 2: contact pairs whose relation to the red ones is the relation type selected in dropdown (G) are drawn in blue.

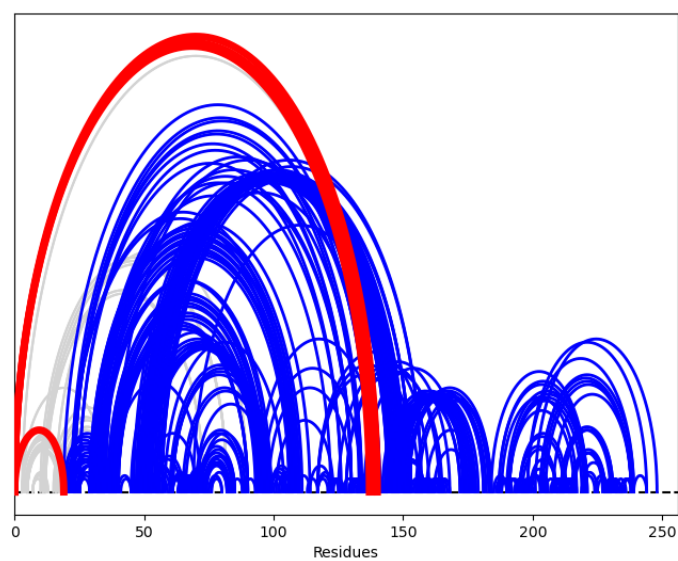


Figure 6: Combined local CT plot showing both the red (residue-anchored) and blue (relation-selected) curves, allowing visual identification of local topological relationships around the selected residue.

## 2.2 Single-File Analysis Protocol

The Single-File Analysis tab gives a global topological overview of one structure: a relations matrix, the corresponding circuit (arc) representation, an optional scalar **Folding Score** per chain, and direct recolouring of the structure in the PyMOL viewer.

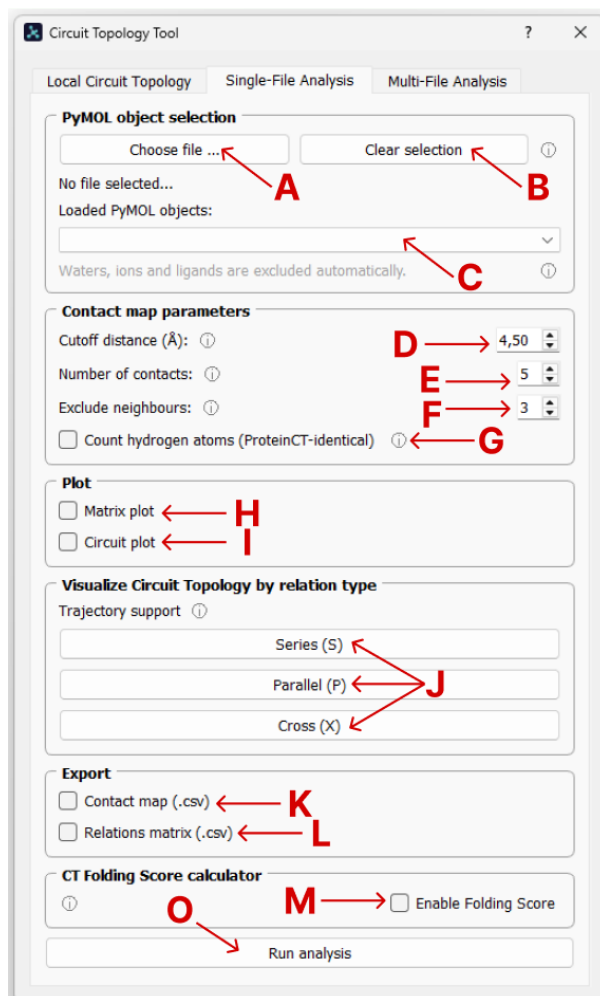


Figure 7: The Single-File Analysis tab, labelled A–M: object selection (A–C), contact-map parameters (D–F) (the *Count hydrogen atoms* checkbox sits below (F)), plot toggles (G) matrix and (H) circuit, the *Visualize Circuit Topology by relation type* buttons (I) (Series, Parallel, Cross), export checkboxes (J–K), the Folding Score toggle (L), and the run button (M).

The **Matrix plot** (G) renders the relations matrix, in which each cell shows the relation type between two contact pairs. For *single-chain* structures the matrix uses the standard S / P / X types together with their inverse and concerted variants. For *multi-chain* structures the whole model is analysed at once: inter-chain contact pairs are included, S, P and X are assigned only to two intra-chain contact pairs of the same chain, and pairs involving an inter-chain contact pair fall into the Independent (I), Tandem (T) or Loop (L) types visible in the legend of Figure 9. The **Circuit plot** (H) draws every contact pair as an arc above the residue index; for multi-chain inputs each chain is rendered in a different colour (Figure 8).

## Single-File Analysis Guide

1. **Load the protein (A–C):** load a `.pdb` or `.cif` file, or pick an existing PyMOL object from the dropdown.
2. **Define contact parameters (D–F):** cutoff distance, number of contacts and excluded neighbours — the same parameters, ranges and defaults as on the LCT tab, plus the **Count hydrogen atoms (ProteinCT-identical)** checkbox.
3. **Select plots:** tick **Matrix plot (G)** for the relations matrix and **Circuit plot (H)** for the arc diagram.
4. **(Optional) Folding Score:** tick **Enable Folding Score (L)** to compute a per-chain CT Folding Score (Section 2.2.2).
5. **Export data:** tick **Contact map (.csv) (J)** and/or **Relations matrix (.csv) (K)**, then choose an output directory in the widgets that appear.
6. **Run:** click **Run analysis (M)**.

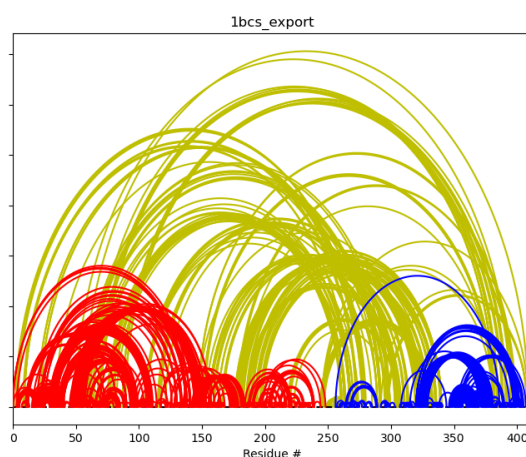


Figure 8: Circuit (arc) plot produced by the Single-File Analysis tab for the multi-chain object `1bcs_export`. Each chain is drawn in a distinct colour; arcs spanning chains are inter-chain contact pairs, which are included in whole-model analysis.

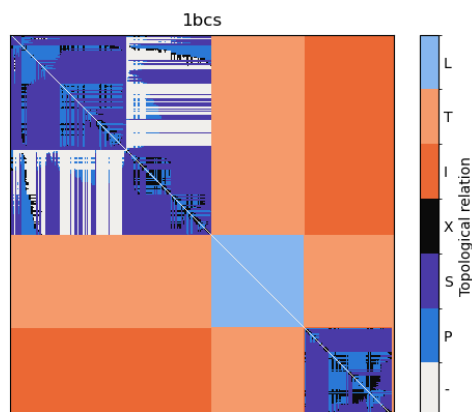


Figure 9: Relations-matrix plot for the same multi-chain object. The legend uses the multi-chain relation types (P, S, X, I, T, L), which appear only for multi-chain proteins. For single-chain inputs the legend reduces to the standard single-chain relation types.

### 2.2.1 Visualizing Relation Types on the 3D Structure in PyMOL

The **Visualize Circuit Topology by relation type** panel (Figure 7, button row **(I)**) recolours the structure in the PyMOL viewer according to how densely each residue participates in a given relation type. Pressing **Series (S)**, **Parallel (P)** or **Cross (X)** runs the analysis for each chain of the selected object (intra-chain contact pairs only) and then, per chain, calls

```
get_relation_type_vector(mat, index, relation_type, numbering)
color_by_relation(molecule_name, relation_vector, numbering,
    relation_type)
```

`get_relation_type_vector` returns one value per residue that counts how many relations of the selected relation type that residue takes part in. The value is aligned element-for-element with the residue numbering.

`color_by_relation` writes the counts into the B-factor column and colours the residues by them.

#### Turning counts into colours

Residues that take part in *no* relation of the selected type are white. Every other residue is placed in one of five levels, from pale tint at lowest to full saturation at highest.

The level boundaries are the *quantiles* of the non-zero counts. This accounts for the heavily right-skewed nature of the residue counts. Spread evenly from zero to the maximum, a few outliers would consume the whole scale — in chain A of **1bcs** the Series counts reach 1900, so a residue taking part in 50 relations would sit 2.6% along a white-to-colour ramp and render as white, indistinguishable from a residue taking part in none. Quantile levels guarantee that every shade is used, so low activity reads as low activity rather than as absence.

## Trajectories

The buttons act on whichever object is selected in the dropdown. If that object is a trajectory with more than one state, *every* state is coloured: the analysis runs for each frame and the frames are merged into a single multi-state object named `<object>_topo` that you can scrub through with PyMOL's frame controls. Every frame is measured before any frame is coloured, so the whole trajectory shares one scale and one legend — otherwise a residue would change colour between frames merely because that frame's distribution shifted, which is exactly the sort of thing an animation makes look real. For long trajectories this takes a while and uses significant memory. Non-polymer atoms are excluded before the frames are split, so solvent never enters the calculation.

**Warnings you may see.** Two limitations inherited from ProteinCT are reported at runtime instead of silent ignore. Contact detection matches on the residue number alone, so where **insertion codes** are in use the atoms of an inserted residue are attributed to the first residue with that number. Neighbour exclusion is applied by position in the residue list rather than by residue number, so residues flanking a **chain break** are treated as sequence neighbours. A representative log for the three-chain structure `1bcs` looks like this:

```
INFO - Chains found:  ['A', 'B', 'C']
WARNING - 1bcs_topo_A: 4 residue number(s) occur more than once, i.e. insertion
codes are in use (23, 58, 76, 112).
WARNING - 1bcs_topo_A: 2 chain break(s) in the residue numbering (11->14,
134->137).
INFO - Analysing 1bcs_topo, chain A ...
WARNING - No contacts found for chain C. Skipping visualization...
INFO - Coloured '1bcs_topo' by S relations. '1bcs' keeps its original
B-factors.
```

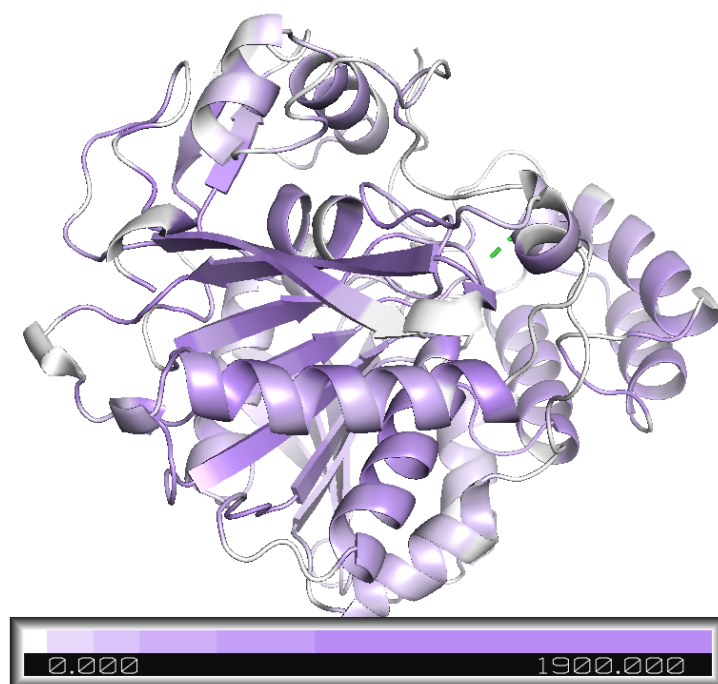


Figure 10: Relation type **Series (S)** visualized for 1bcs. Chain C yielded no contacts and was skipped.

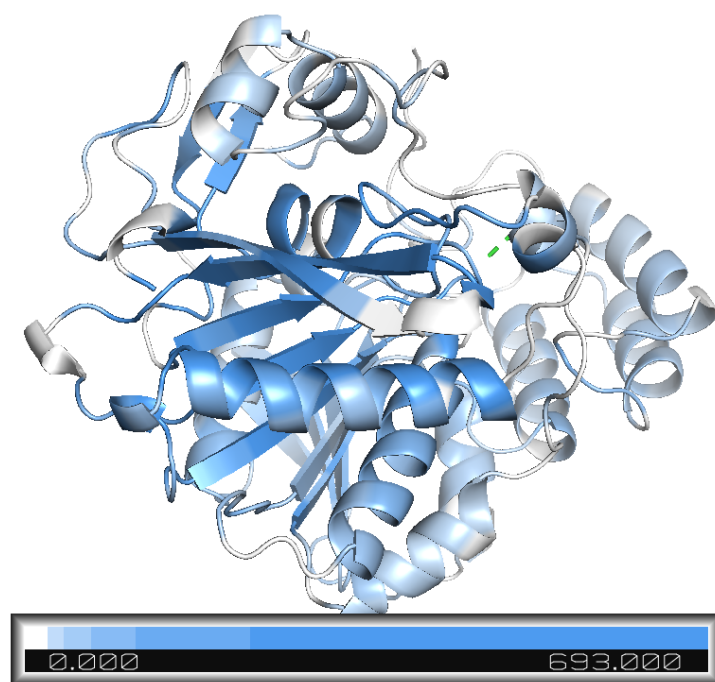


Figure 11: Relation type **Parallel (P)** visualized for 1bcs.

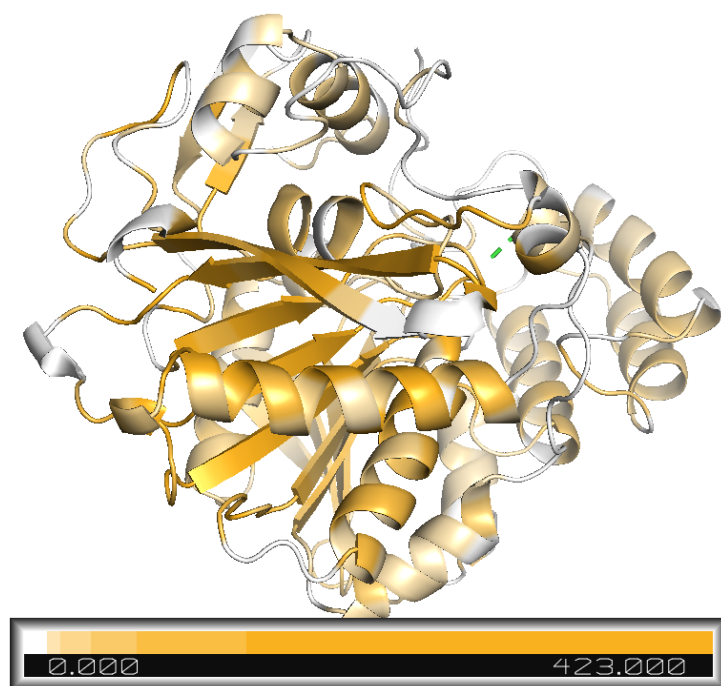


Figure 12: Relation type **Cross (X)** visualized for 1bcs. Comparing the three figures shows which regions are densely involved in each relation type.

### 2.2.2 CT Folding Score

Ticking **Enable Folding Score (L)** reduces the contact map and relations matrix to a single number per chain: the sum of the mean per-residue participation counts in Parallel, Series and Cross relations, computed by `get_folding_score` from the intra-chain contact pairs of that chain. Larger values indicate a contact network in which the average residue participates in more topological relations. The score builds on the circuit-topology-based fold analysis framework of Akulov *et al.* [3] and gives a quick way to compare conformations without inspecting full plots.

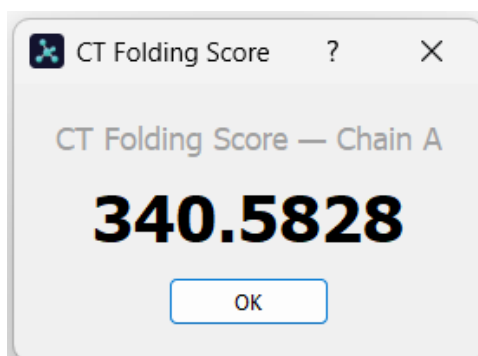


Figure 13: CT Folding Score for protein 1a6n.

## 2.3 Multi-File Analysis Protocol

The Multi-File Analysis tab batch-processes a set of structures: either a directory of .pdb/.cif files, or a trajectory loaded as a topology .pdb/.cif plus an .xtc trajectory file.

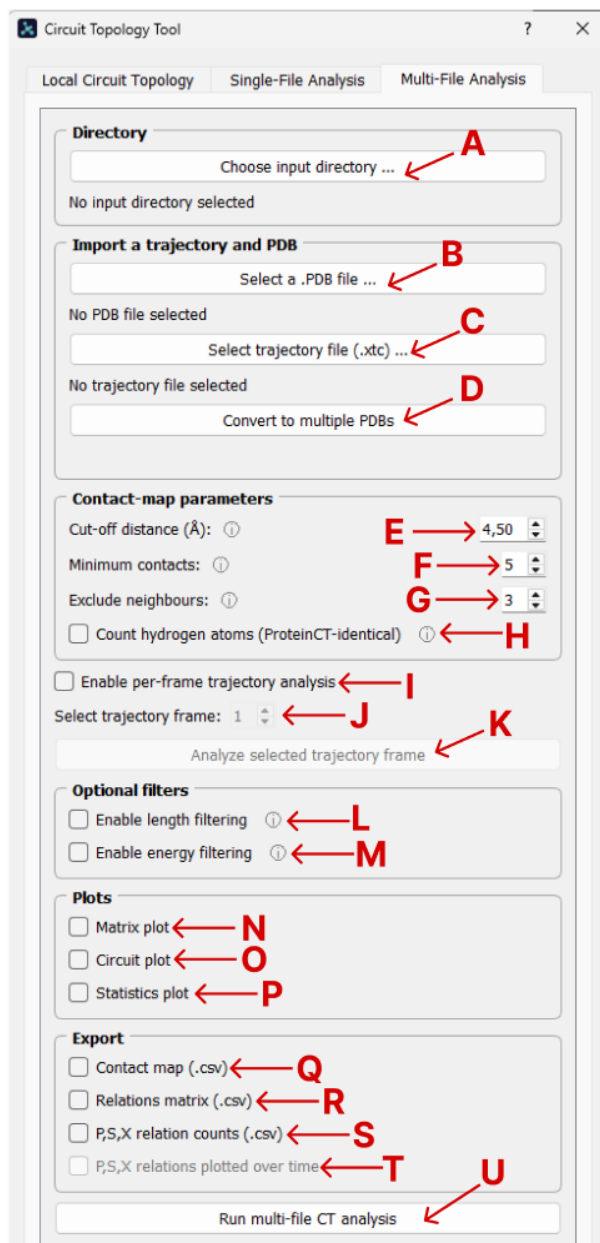


Figure 14: The Multi-File Analysis tab, labelled A–T: directory input (A), trajectory import (B–C), trajectory-to-PDB conversion (D), contact-map parameters (E–G) (the *Count hydrogen atoms* checkbox sits below (G)), per-frame analysis (H–J), optional filters (K) length and (L) energy, plot toggles (M) matrix, (N) circuit, (O) statistics, export checkboxes (P–S), and the run button (T).

### Input Modes

Two input modes are supported. Either

- point at a **directory** of structures via **Choose input directory ...** (A), or

- import a topology and trajectory pair: **Select a .PDB file ... (B)** and **Select trajectory file (.xtc) ... (C)**.

With a topology and trajectory loaded, **Convert to multiple PDBs (D)** writes each frame to its own `.pdb` file, turning the trajectory into a directory you can then point at for a regular multi-file run. This is optional: a loaded trajectory can be analysed directly, frame by frame, without exporting anything.

## Per-Frame Analysis

Ticking **Enable per-frame trajectory analysis (H)** enables the **Select trajectory frame** spinner (**I**) and the **Analyze selected trajectory frame** button (**J**), which processes only the chosen frame. Both stay greyed out until the checkbox is ticked. As an alternative, the imported trajectory object can be selected on the Single-File Analysis tab and stepped through with PyMOL's own frame controls.

## Optional Filters

- **Enable length filtering (K)**: keeps only contacts whose sequence separation matches the chosen comparator — **equality (=)**, **less than (<)** or **greater than (>)** — against a **Distance** threshold. Ticking the box reveals the threshold spinner and the comparator dropdown.
- **Enable energy filtering (L)**: retains only **Attractive / stabilising (+)** or only **Repulsive / destabilising (-)** contacts. Ticking the box reveals the mode dropdown.

### Filter caveats

Both filters were designed for single-chain proteins and are skipped automatically when the input contains multiple chains — the plugin reports which file and which filter it skipped. They also have no effect during *per-frame trajectory analysis*, which runs the unfiltered pipeline on the chosen frame. Aggressive length-filter thresholds can produce empty contact maps.

## Plots and Exports

Three plot checkboxes are available:

- **Matrix plot (M)** and **Circuit plot (N)** — the same outputs as the Single-File tab, generated for every input file.
- **Statistics plot (O)** — a two-panel figure: the fraction of P + X relations, i.e. the share of contact-pair relations that are Parallel or Cross (non-Series), against distance from the matrix diagonal — the quantity ProteinCT calls the “entangled” fraction, which describes relation types rather than geometric entanglement — beside a pie chart of the Parallel, Series and Cross proportions.

Three export checkboxes write CSV files to the chosen output directory: the residue contact map (**P**), the relations matrix (**Q**), and the P, S and X relation counts per structure (**R**) (**P,S,X relation counts (.csv)**, one row per file or frame). Ticking it

enables the fourth checkbox, **P,S,X relations plotted over time (S)**, which plots the three counts across the input set. That is only meaningful for a genuine trajectory, i.e. when each input file is a consecutive frame of the same structure. As on the other tabs, the output-directory widgets appear as soon as any export is ticked.

### Multiple-File Analysis Guide

1. **Load the input:** pick an input directory (**A**) of .pdb/.cif files, or load a topology and trajectory pair (**B–C**).
2. **(Optional) Convert the trajectory to PDBs (D)** if you want the individual frames as standalone files.
3. **Define contact parameters (E–G):** cut-off distance, minimum contacts, exclude neighbours, and whether hydrogen atoms are counted (off by default). For a molecular dynamics trajectory this setting decides whether the all-atom frames are analysed on their heavy atoms (default) or on every atom.
4. **(Optional) Per-frame analysis (H–I):** tick **Enable per-frame trajectory analysis** and choose the frame index.
5. **(Optional) Apply filters (K–L):** enable length and/or energy filtering, subject to the caveats above.
6. **Pick plots (M–O):** matrix, circuit, statistics.
7. **Pick exports (P–R):** contact map, relations matrix, P, S, X relation counts, and optionally (**S**) for the time series.
8. **Run:** click **Run multi-file CT analysis (T)** to process the whole input set, or **Analyze selected trajectory frame (J)** to process just the chosen frame.

## References

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- [1] Moes, D., Banijamali, E., Sheikhhassani, V., Scalvini, B., Woodard, J., Mashaghi, A. (2022). *ProteinCT: An implementation of the protein circuit topology framework*. MethodsX, open access (PII S2215016122002400). Source code: [https://github.com/circuittopology/circuit\\_topology](https://github.com/circuittopology/circuit_topology).
- [2] Mashaghi, A., Van Wijk, R.J., Tans, S.J. (2014). *Circuit topology of proteins and nucleic acids*. Structure 22(9), 1227–1237. doi:10.1016/j.str.2014.06.015.
- [3] Akulov, V., Jiménez Panizo, A., Estébanez-Perpiñá, E., van Noort, J., Mashaghi, A. (2025). *Phosphorylation regulated conformational diversity and topological dynamics of an intrinsically disordered nuclear receptor*. The Journal of Physical Chemistry B 129(30), 7719–7730. doi:10.1021/acs.jpcb.5c03257.
- [4] Heidari, M., Moes, D., Schullian, O., Scalvini, B., Mashaghi, A. (2022). *A topology framework for macromolecular complexes and condensates*. Nano Research 15(11), 9809–9817. doi:10.1007/s12274-022-4355-x.